MILITARY SPECIFICATION
FOR
TEST SET, MISSILE ELECTRICAL SYSTEM
MARK 363 MOD 3

This specification has been approved by
The Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 This specification covers a Missile Electrical System Test Set Mk 363 Mod 3 for use in testing ASROC (Anti-Submarine Rocket) weapon system electrical circuits, hereafter referred to as the "Test Set."

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification, to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-621	Boxes, Wood, Nailed and Lock Corner

Military

MIL-P-116	Reservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

FSC 4935

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DRAWINGS

Bureau of Naval Weapons

BUWEPS LD 477269	Test Set, Missile Electrical System, Mark 363 Mod 3 and all documents listed thereon

PUBLICATIONS

Bureau of Naval Weapons

BUWEPS OD 13776

Revision 1 CHANGE 3	Test Set, Missile Electrical System, Mark 363 Mod 3 Factory Acceptance Test

(Copies of specifications, standards, drawings, and publications required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Description. The Test Set simulates an entire missile electrical system. This includes basic functions of the Range and Airframe Separation Programmer and Selection of Initial Search Depth and Floor Depth, which are functions of the torpedo. In service use, the Test Set is connected to the launcher to provide to the weapon system responses which will test the system circuits, and also check the system for safety.

3.2 General Requirements

3.2.1 Materials and Treatments of Materials and Parts. All materials, treatments, and parts used in the manufacture of the Test Set shall be of such quality as to enable the Test Set to meet the requirements herein.

3.2.2 Gages. The supplier shall provide those gages necessary and adequate to assure that the Test Set will meet dimensions and requirements shown on the applicable documents listed in LD 477269. When requested by the supplier, the procuring activity may furnish existing, pertinent drawings of Naval final inspection gages for use only in the designing of the suppliers gages. This action does not relieve the supplier of the responsibility for designing and manufacturing the gages needed for the fulfillment of the contract.

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3.2.3 Requests for Deviation. Requests for deviation from this specification, applicable drawings, specifications, publications, materials, and processes specified herein, shall be submitted to the Procuring Activity. The deviation shall have been authorized by the Bureau of Naval Weapons or by a Naval Activity delegated such authority, prior to the production of the affected item or prior to the performance of a particular operation on the affected item. The request shall define the suggested deviation in sufficient detail to permit the Procuring Activity to evaluate its effect on performance and reliability, and the reasons for the suggested deviation shall be stated.

3.2.4 Reproduction Sample. Unless otherwise specified in the contract or order, the preproduction sample shall:

a. Be manufactured using procedures proposed for production.

b. Be tested as specified in section 4 herein and is for the purpose of determining that, prior to starting production, the supplier's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.3 Performance and Product Characteristics

3.3.1 Operation

a. The Test Set POWER lamp shall light when 28 plus or minus 3 volts DC is applied across terminals k (+) and h (-) of panel connector P3.

b. All PRESS-TO-TEST lamps on the panel shall light when pressed.

c. The ISD, FLOOR, and POWER-ON lamps shall not be illuminated with CB1 in the TRIP position.

3.3.2 Voltage Safety Requirements

3.3.2.1 Safe Voltage. Meter Ml shall indicate within the GREEN area of the SAFE VOLTS scale when 0.5 volts plus or minus 10 percent AC or 0.5 volts plus or minus 10 percent DC of either polarity is applied across connector P2 pins as follows:

a. Across A and B with FUNCTION SELECTOR in 1 & COM position.

b. Across C and B with FUNCTION SELECTOR in 2 & COM position.

c. Across A and C with FUNCTION SELECTOR in 1 & 2 position.

3.3.2.2 Unsafe Voltage. Meter Ml shall indicate within RED area of the SAFE VOLTS scale when 1 volt plus or minus 10 percent AC or 1 volt plus

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or minus 10 percent DC of either polarity is applied across connector P2 pins as follows:

a. Across A and B with FUNCTION SELECTOR in 1 & COM position.

b. Across C and B with FUNCTION SELECTOR in 2 & COM position.

c. Across A and C with FUNCTION SELECTOR in 1 & 2 position.

3.3.3 Meter Circuit Protection. With 28 plus or minus 3 volts DC applied to pins k (+) and h (-) of connector P3, the reset meter lamp shall light and latching relay K6 shall operate and open the safe voltage protection circuit and the relay shall remain latched and the lamp shall remain lighted as long as 3.3 plus or minus 1.0 volts AC or 3.5 plus or minus 1.0 DC of either polarity is present across pins A and B of P2 with FUNCTION SELECTOR in 1 & COM position; across pins C and B of P2 with FUNCTION SELECTOR in 2 & COM position; or across pins A and C of P2 with FUNCTION SELECTOR in 1 & 2 position.

3.3.4 Firing Lines Circuit Safe Indication. With 28 plus or minus 3 volts DC applied to pins k (+) and h (-) of connector P3, and with Ml meter circuit calibrated, resistance of 2 ohms plus or minus 1 percent between connector pins A of P2 and T of P3, C of P2 and T of P3, and B of P2 and T of P3, shall indicate within the GREEN area of the CIRCUIT SAFE scale of meter Ml when the FUNCTION SELECTOR is placed in 1 & GRD, 2 & GRD, or COM & GRD positions.

3.3.5 Firing Lines Circuit Unsafe Indication. With 28 plus or minus 3 volts DC applied to pins k (+) and h (-) of connector P3, and with Ml meter circuit calibrated, resistance of 4 ohms plus or minus 1 percent between connector pins A of P2 and T of P3, C of P2 and T of P3, B of P2 and T of P3, shall indicate within the RED area of the CIRCUIT SAFE scale of meter Ml when the FUNCTION SELECTOR is placed in 1 & GRD, 2 & GRD or COM & GRD positions.

3.3.6 Battery Ground Test

3.3.6.1 Battery Line Circuit Ground Indication. With 28 plus or minus 3 volts DC applied to pins k (+) and h (-) of connector P3, calibrate meter Ml. With the FUNCTION SELECTOR in BAT GRD position, resistance of zero (0) ohms (short) between connector pins R and W of P3 shall indicate at least three-fourths (3/4) full scale deflection of meter Ml.

3.3.6.2 Battery Line Circuit Partial Ground Indication. With 28 plus or minus 3 volts DC applied to pins k (+) and h (-) of connector P3, calibrate meter Ml. With the FUNCTION SELECTOR in RAT GRD position, resistance of 3 ohms plus or minus 1 percent between connector pins R and W of P3 shall indicate in the "CK" area of meter Ml scale.

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3.3.6.3 Battery Line Circuit No Ground Indication. With 28 plus or minus 3 volts DC applied to pins k (+) and h (-) of connector P3, calibrate meter Ml. With the FUNCTION SELECTOR in RAT GRD position, resistance of 100 ohms plus or minus 1 percent between connector pins R and W of P3 shall indicate in the GREEN area of the SAFE VOLTS scale.

3.3.7 Firing Voltage

3.3.7.1 Battery Position. With the FUNCTION SELECTOR in START position, MOTOR START-BATTERY START switch in BATTERY START, apply 20 volts AC, plus or minus 10 percent, so phased across pins H, R and J of connector P3 as to Provide 20 volts AC plus or minus 10 percent between pins H and R, and 40 volts AC plus or minus 10 percent between pins H and J, the THERMAL START lamp shall light. Meter Ml shall indicate within the "CK" area with the 20 V LOAD-40 V LOAD switch in the 20 V LOAD position and within the "CK" area with the 20 V LOAD-40 V LOAD switch in the 40 LOAD position when voltages of 18 and 22 plus or minus 1 percent AC, and 36 and 44 plus or minus 1 percent AC are applied. The THERMAL START lamp shall extinguish after removal of the AC voltages and operation of the RESET switch.

3.3.7.2 Motor Position

a. With the FUNCTION SELECTOR in START and with MOTOR START-BATTERY START stitch in MOTOR START position, and no voltage applied to connector P3 pins, H, R and J, there shall be no continuity between Pins h and g of P3.

b. With AC voltages applied as specified in 3.3.7.1, THERMAL START lamp shall light immediately and after a delay of 1.00 plus or minus 0.30 seconds, there shall be continuity between pins h and g of connector P3.

c. With AC voltages applied as specified in 3.3.7.1, and with 20 volts AC plus or minus 10 percent so phases across pins A, B and C of connector P2 as to provide 20 volts AC plus or minus 10 percent between pins A and B, and 40 volts AC plus or minus 10 percent between pins A and C, the START lamp shall light. Meter Ml shall indicate within the "CK" area with the 20 V LOAD- 40 V LOAD switch in the 20 V LOAD position and within the "CK" area with the 20 V LOAD - 40 V LOAD switch in the 40 V LOAD position when voltages of 18 and 22 plus or minus 1 percent AC and 36 and 44 plus or minus 1 percent AC are applied. The START and THERMAL START lamps shall extinguish after removal of the AC voltages and operation of the RESET switch.

3.3.8 Simulated Missile Identification Functions

3.3.8.1 DC Interlock Simulation. With no voltage applied to pins k (+) and h (-) of connector P3, there shall be continuity between pins L, h and Z of connector P3, when DC TORP switch is in DC position. The DC INTLK switch in the open position shall disrupt continuity between pins L and Z and between L and h of connector P3.

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3.3.8.2 Misfire Simulation. There shall be no continuity between pins e and f of connector P3 with DUD switch in normally open position. There shall be continuity between pins e and f when DUD switch is operated to closed position.

3.3.9 Torpedo Simulation

3.3.9.1 28 Volt Supply Ground. With DC-TORP switch in TORP position, there shall be continuity between pins h and T and h and X of connector P3.

3.3.9.2 ISD Selector. With the FUNCTION SELECTOR in START position and with the DC-TORP switch in TORP position, apply 28 volts plus or minus 1 percent between pins m (+) and h (-) of connector P3 and apply 28 volts plus or minus 1 percent between pins n (+) and h (-) of connector P3. The ISD indicator lamps shall illuminate, indicating the relative position of the ISD (S2) stepping switch. When the ISD switch is remotely rotated from 50 to 900, the voltages between pins M (+) and h (-) of P3 increases 4 volts with each increase in the ISD switch position up to a maximum of 24 volts as shown in Table 1. The proper ISD indicator lamp shall illuminate for each position.

TABLE 1
ISD SELECTOR OUTPUT VOLTAGES

ISD Switch Position	Voltage Readings - M (+) to h (-)
50	4 volts DC ± 5%
150	8 volts DC ± 5%
250	12 volts DC ± 5%
450	16 volts DC ± 5%
650	20 volts DC ± 5%
900	24 volts DC ± 5%

3.3.9.3 FLOOR Selector. With the FUNCTION SELECTOR in START position and with the DC-TORP switch in TORP position, apply 28 volts plus or minus 1 percent between pins m (+) and h (-) of connector P3 and apply 28 volts plus or minus 1 percent between pins p (+) and h (-) of connector P3. The FLOOR indicator lamps shall illuminate, indicating the relative position of the FLOOR (S3) stepping switch. When the FLOOR switch is remotely rotated from 150 to 900, the voltages between pins N (+) and h (-) of connector P3 increases 5.6 volts with each increase in the FLOOR switch position up to a maximum of 28 volts as shown in Table 2. The proper FLOOR indicator lamp shall illuminate for each position.

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TABLE 2
FLOOR SELECTOR OUTPUT VOLTAGES

FLOOR Switch Position	Voltage Readings - N (+) to h (-)
150	5.6 volts DC ± 10%
250	11.2 volts DC ± 10%
450	16.8 volts DC ± 10%
650	22.4 volts DC ± 10%
900	28.0 volts DC ± 10%

3.3.10 Readout Timing Requirements

3.3.10.1 DMma Readout Timing. With 28 plus or minus 3 volts DC applied across pins k (+) and h (-), and 12.750 plus or minus 0.250 volts DC across pins S (+) and T (-) of connector P3; with FUNCTION SELECTOR in START, MOTOR START-BATTERY START in MOTOR START, and with DMma-OFF-T4 in DMma position, DMma readout times on Test Set clock shall be as shown in Table 3, after application of presaturation, thermal battery start, set-in, and motor start voltages in the following sequence:

a. Presaturation voltage of 10.000 plus or minus 0.010 volts DC applied to pin W (-) and to pin F (+) of connector P3 for not less than 2 seconds.

b. Thermal battery start voltage of 20 volts AC plus or minus 10 percent across pins H and R, and across pins R and J, so phased as to provide 40 volts AC plus or minus 10 percent across pins H and J, for not less than 1 second.

c. Set-in voltages as shown in Table 3 for 100 plus or minus 0.1 milliseconds across pins F (+) and W (-) of connector P3, after continuity appears between pins h and g. Rise time of the 100 millisecond pulse shall not be greater than 150 microseconds and fall time not greater than 250 microseconds. Pulse width shall be measured at the -25 volt level with a -50 volt set-in. Switching time from presaturation to set-in voltage shall not be greater than 1 millisecond.

d. Motor start voltage of 20 volts AC plus or minus 10 percent across pins A and B of P2, and across pins B and C of P2, so phased as to provide 40 volts AC plus or minus 10 percent across pins A and C of P2, and applied coincidentally with the end of the 100 millisecond pulse.

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TABLE 3
DMma READOUT TIMES

Set-in Volts DC	Mk 363 Timer Readings
.25.000 ± 0.025	1.00 ± 0,100 seconds
-50.000 ± 0.050	2.00 ± 0.200 seconds
-100.000 ± 0.100	4.00 ± 0.400 seconds

3.3.10.2 T4 Readout Timing. With 28 plus or minus 3 volts DC applied across pins k (+) and h (-) and 12.750 plus or minus 0.250 volts DC across pins S (+) and T(-) of connector P3; with FUNCTION SELECTOR in START, MOTOR START-BATTERY START in MOTOR START, and with DMma-OFF-T4 in T4 position, T4 readout times shall be as shown in Table 4, after the sequence of voltages as specified in 3.3.10.1 (a.) through (d.), except that in step (a.) the pre-saturation and set-in voltages shall be applied through pins C (+) and W (-) of connector P3.

TABLE 4
T4 READOUT TIMES

Set-in Volts DC	Mk 363 Timer Readings
-200000 ± 0.020	10.00 ± 1.000 seconds
-40.000 ± 0.040	20.00 ± 2.000 seconds
-80.000 ± 0.080	40.00 ± 4.000 seconds

3.3.11 Range and Airframe Separation Simulation

3.3.11.1 Operating Voltages. With 12.750 plus or minus 0.250 volts DC applied across pins S (+) and T (-) of connector P3, the following voltages shall be present at the designated pins of connector P3:

a. 12.750 plus or minus 0.250 volts DC between pins A (+) and V (-).

b. 10.000 plus or minus 0.010 volts DC between pins d (+) and T (-)

c. 2.500 plus or minus 0.010 volts DC between pins B (+) and T (-)"

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d. 0.200 plus or minus 0.001 volts DC! between pins E (+) and T (-).

e. 28 plus or minus 3 volts DC between pins c (+) and T (-).

3.3.11.2 DMma Output. Elapsed time from the instant the 2.500 voltage is applied in (c) below until the voltage across pins G (+) and W (-) reaches 4.O plus or minus 0.5 volts DC, shall be as shown in Table 5 after the application of voltages in the sequence specified in steps (a.) through (c.) below. The variation of plus voltage between pins G (+) and W (-) shall not exceed 0.500 volts peak after application of the 100 millisecond pulse in step (c.) and prior to completion of the time periods specified in Table 5.

a. 10.000 plus or minus 0.010 volts DC to pins W (-) and F (+) of P3 for not less than 2 seconds.

b. Set-in voltages across pins F (+) and W (-) as specified in Table 5 for 100 plus or minus 0.1 milliseconds. Rise time of the 100 millisecond pulse shall not be greater than 150 microseconds and fall time not greater than 250 microseconds. Pulse width shall be the width measured at the -25 volt level with a -50 volt set-in. Switching time from presaturation to set-in voltage shall not be greater than 1 millisecond.

c. 2.500 plus or minus 0.010 volts DC across pins F (+) and W (-).

TABLE 5
DMma READOUT TIMES

Set-in Volts DC	Readout Seconds
-25.000 ± 0.020	1.00 ± 0.020 seconds
-50.000 ± 0.020	2.00 ± 0.020 seconds
-100.000 ± 0.020	4.00 ± 0.020 seconds

3.3.11.3 T4 Output

3.3.11.3.1 Readout at 0.200 Volts. Elapsed time from the instant a 0.200 voltage is applied in (c.) below until the voltage across pins D (+) and W (-) of P3 reaches 4.0 plus or minus 0.5 volts DC, shall be as shown in Table 6 after the application of voltages in the sequence specified in steps (a.) through (c.). The variation of plus voltage between pins D (+) and W (-) of P3 shall not exceed 0.500 volts peak after application of the 100 millisecond pulse in step (c.) and prior to completion of the time periods specified in Table 6.

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a. 10.000 plus or minus 0.010 volts DC to pin W (-) and to pin C (+) of connector P3 for not less than 2 seconds.

b. Set-in voltages across pins C (+) and W (-) as specified in Table 6 for 100 plus or minus 0.1 milliseconds. Rise time of the 100 millisecond pulse shall not be greater than 150 microseconds and fall time not greater than 250 microseconds. Pulse width shall be measured at the -25 volt level with a -50 volt set-in. Switching time from presaturation to set-in voltage shall not be greater than 1 millisecond.

c. 0.200 plus or minus 0.001 volts DC across pins C (+) and W (-).

TABLE 6
T4 READOUT TIMES

Set-in Volts DC	Readout Seconds
-20.000 ± 0.020	10.00 ± 0.500 seconds
-40.000 ± 0.020	20.00 ± 0.500 seconds
-80.000 ± 0.020	40.00 ± 0.500 seconds

3.3.11.3.2 Readout at 2.500 Volts. Elapsed time from the instant a 2.500 voltage is applied in (c.) below until the voltage across pins D (+) and W (-) of P3 reaches 4.0 plus or minus 0.5 volts DC shall be 2.00 plus or minus 0.10 seconds after the application of voltages in the sequence specified in steps (a.) through (c.) below. The variation of plus voltage between pins D (+) and W (-) of P3 shall not exceed 0.500 volts peak after application of the 100 millisecond pulse in step (c.) and prior to completion of the the period of 2.00 plus or minus 0.10 seconds.

a. 10.000 plus or minus 0.010 volts DC! to pin W (-) and to pin C (+) of connector P3 for not less than 2 seconds.

b. A set-in voltage of -50.000 plus or minus 0.020 volts DC across pins C (+) and W (-) for 100 plus or minus 0.1 milliseconds. Rise time, fall time, pulse width, and switching time requirements shall be as specified in 3.3.11.3.1 (b.).

c. 2.500 plus or minus 0.010 volts DC across pins C (+) and W (-).

3.4 Environmental Requirements

3.4.1 High Temperate. The Test Set shall withstand a temperature of plus 160 plus 5 minus 0 degrees F for not less than 8 hours, and thereafter meet all operational requirements specified herein while the Test Set is stabilized at plus 120 plus 5 minus 0 degrees F.

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3.4.2 Low Temperature. The Test Set shall withstand a temperature of minus 65 plus 0 minus 5 degrees F for not less than 8 hours and thereafter meet all operational requirements specified herein while the Test Set is stabilized at plus 35 plus 0 minus 5 degrees l?.

3.4.3 Humidity. The Test Set shall withstand relative humidity up to 95 percent, as specified in Procedure 1, Specification MIL-E-5272, and thereafter meet performance and product characteristics specified herein.

3.4.4 Fungus. The Test Set shall resist fungus during storage, as specified in Fungus Resistance Tests, Procedure 1, Specification MIL-E-5272, and thereafter meet performance and product characteristics specified herein.

3.4.5 Sand and Dust. The Test Set, with cover removed, shall withstand sand and dust conditions as specified in Procedure III, Specification MIL-E-5272, and thereafter meet performance and product characteristics specified herein.

3.4.6 Salt Spray. The Test Set, with cover removed, shall withstand salt spray conditions specified in Procedure 1, Specification MIL-E-5272, and thereafter meet performance and product characteristics specified herein.

3.4.7 Rain. The Test Set, with cover removed, shall withstand rain conditions as specified in Procedure II, Specification MIL-E-5272, and thereafter meet performance and product characteristics specified herein.

3.4.8 Shock. The Test Set shall withstand shock tests as specified for Equipment Operation, Procedure V, Specification MIL-E-5272, and thereafter meet performance and product characteristics specified herein.

3.4.9 Vibration. The Test Set shall withstand vibration up to 33 CPS as specified for Type 1 equipment in Standard MIL-STD-167, and meet all operational requirements specified herein while subjected to this environment.

3.5 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine screw assemblies, welding, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own facilities or any commercial laboratory

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acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspections. Inspection of the Test Set shall be classified as follows:

a. Preproduction inspection
b. Operational Tests inspection
c. Periodic production inspection

4.2.1 Preproduction Sample. A preproduction sample of one Test Set manufactured in accordance with section 3 shall be subjected to the preproduction tests detailed in (paragraphs 4.4.1 through 4.4.9) at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the Test Set by the supplier, prior to the approval of the preproduction sample shall be at the supplier's risk.

4.2.2 Periodic Production Sampling. Periodic production samples shall be selected from units that have satisfactorily passed the quality conformance inspection of OD 13776. These samples shall be subjected to the additional quality conformance inspection of (paragraphs 4.4.1 through 4.4.9) at an activity designated by the procuring actitity. Periodic production sampling shall be as follows:

Step 1. Upon initiation of production randomly select one unit from each group of ten units. Upon satisfactory compliance of samples selected from two consecutive groups, shift to Step 2.

Step 2. Randomly select one unit from each group of twenty-five units. Upon satisfactory compliance of sample selected from two consecutive groups, shift to Step 3.

Step 3. Randomly select one unit from each group of fifty units. Continue with Step 3 for the balance of the contract or until a defective unit is found.

When a defective unit is found in any group, acceptance of all units after the last group of accepted units shall be withheld and the contractor shall:

a. Determine the cause of the defect.

b. Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect. Return to Step 1 and proceed as indicated when inspection is resumed.

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4.2.3 Warning to Contractor. Construction of the Missile Test Set in accordance with drawings listed on BUWEPS LD 477269 shall not be construed as a waiver of any of the requirements of this and other applicable specifications.

4.3 Tests

4.3.1 Operational Tests. Each Test Set delivered under contract shall be subjected to the tests and examinations listed in OD 13776 Revision 1, Change 3. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.3.2 Test Conditions. Unless otherwise specified, the Test Set shall be subjected to acceptance tests under the following conditions:

a. Temperature - plus 65 to plus 95 degrees F.
b. Altitude - normal ground atmospheric pressure.
c. Humidity - room ambient to 95 percent relative.
do Vibration - none

4.3.3 Test and Inspection Equipment and Facilities. The manufacturer shall furnish and maintain all necessary test equipment, facilities and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity, and, unless definite requirements are specified, they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.3.4 Test Procedure. The Test Set shall meet the requirements of 3.3 when tested as specified in OD 13776 Revision 1, Change 3.

4.4 Preproduction and Periodic Production Tests. The preproduction and periodic samples, after satisfactorily passing the quality conformance inspections of OD 13776 Revision 1, Change 3, shall be subjected to the additional quality conformance inspections (paragraphs 4.4.1 through 4.4.9). Recalibration of the Test Set shall be permitted after having been subjected to the tests of (4.4.1 through 4.4.9) and before being retested for the quality conformance inspections of OD 13776 Revision 1, Change 3.

4.4.1 High Temperature. The Test Set, not energized, and with the cover in place and latched, shall be placed within a suitable test chamber and the chamber temperature raised to plus 160 plus 5 minus 0 degree F. The chamber shall remain at this temperature for not less than eight hours after which the temperature shall be lowered to plus 120 plus 5 minus 0 degrees F, the cover removed and and temperature stabilized to 120 plus 5 minus 0 degrees F for six hours. The Test Set shall then be energized. The Test Set shall meet all the performance and product characteristics of this specification while maintained at the temperature of plus 120 plus 5 minus 0 degrees F.

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4.4.2 Low Temperature. The Test Set, not energized and with the cover in place and latched, shall be placed within a suitable test chamber and the chamber temperature lowered to minus 65 plus 0 minus 5 degrees F. The chamber shall remain at this temperature for not less than eight hours after which the temperature shall be raised to plus 35 plus 0 minus 5 degrees F, the cover removed and the temperature stabilized at plus 35 plus 0 minus 5 degrees F for six hours. The Test Set shall then be energized. The Test Set shall meet all performance and product characteristics of this specification while maintained at the temperature of plus 35 plus 0 minus 5 degrees F.

4.4.3 Humidity. With the cover removed for the first three cycles (24 hours) the Test Set shall be subjected to the humidity test, Procedure I Specification MIL-E-5272. The cover shall then be put in place and latched for the remainder of the tests and thereafter the Test Set shall meet the performance and product characteristics of this specification.

4.4.4 Fungus. The Test Set, with the cover in place and latched, shall be subjected to the fungus test, Procedure I, Specification MIL-E-5272 and thereafter meet the performance and product characteristics of this specification.

4.4.5 Sand and Dust. With exception that the air velocity shall be between 100 and 500 feet per minute, the Test Set, with cover removed, shall be subjected to the sand and dust test, Procedure III, Specification MIL-E-5272 and thereafter meet the performance and product characteristics of this specification.

4.4.6 Salt Spray. With the cover removed for the first phase (48 hours), the Test Set shall be subjected to the salt spray test, Procedure I, Specification MIL-E-5272. The cover shall then be put in place and latched for the remainder of the test and thereafter the Test Set shall meet the performance and product characteristics of this specification.

4.4.7 Rain. The Test Set, with cover removed, shall be subjected to the rain test, Procedure II, Specification MIL-E-5272 and thereafter meet the performance and product characteristics of this specification.

4.4.8 Shock. The Test Set, with the cover in place and latched, shall be subjected to 18 impact shocks of 15g acceleration, each shock impulse having a time duration of 11 plus or minus 1 millisecond. The intensity shall be within plus or minus 10 percent when measured with a filter having a band width of 5 to 100 cycles per second, The maximum acceleration shall be reached in approximately 5-1/2 milliseconds. The shock shall be applied in the following directions:

a. Vertically, 3 shocks in each direction.
b. Parallel to the major horizontal axis, 3 shocks in each direction.

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c. Parallel to the minor horizontal axis, 3 shocks in each direction.

The Test Set shall not suffer damage or subsequently fail to meet the performance and product characteristics of the specification.

4.4.9 Vibration. Subject the Test Set to vibration tests for Type 1 equipment as specified in Standard MIL-STD-167 up to 33 CPS. The Test Set shall meet all operational requirements herein while subjected to this environment.

4.5 Cause for Rejection. Failure of the Missile Test Set to meet any of the requirements of this specification shall be considered cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level A. (See 6.4)

5.1.1.1 Cleaning. The Test Set shall be cleaned as specified in Specification MIL-P-116, Method C-1.

5.1.1.2 Drying. The Test Set shall be dried as specified in Specification MIL-P-116, Method D-4.

5.1.1.3 Unit Packaging. The Test Set shall be individually packaged as specified in Specification MIL-P-116, Method IC-2.

5.1.2 Level C. (See 6.4) Cleaning, drying, preservation and packaging, shall be in accordance with the supplier's commercial practice, when such practices have been proven by testing to conform to all applicable requirements.

5.2 Packing

5.2.1 Level A (See 6.4) One unit package of the Test Set, shall be packed in a nailed wood box conforming to the requirements of Specification PPP-B-621, Class 2, Type 1 and 2 load. The container shall be strapped in accordance with instructions contained in the appendix of the above specification. The pack shall protect the Test Set within the shock requirements of this specification when subjected to the Rough Handling Test of MIL-P-116.

5.2.2 Level C. One or more unit packages shall be packed in accordance with supplier's commercial practice. The pack shall protect the Test Set within the shock requirements of this specification when subjected to the Rough Handling Test of MIL-P-116.

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5.3 Marking

5.3.1 Normal Marking. In addition to the markings required by contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

6. NOTES

6.1 Intended Use. The Test Set is intended to be used in performing a Launcher Safety Test and functionally testing the Missile Electrical System.

6.2 Ordering Data. Procurement documents should specify:

a. Title, date and number of this specification.
b. Preservation, packaging, and packing level.
c. Any exceptions to this specification.

6.3 Definitions

6.3.1 Shock. Shock is a change in motion of a body resulting from application of external force of short duration. Shock is a pulse in units of gravitational acceleration, g, duration in units of time, milliseconds, and a shock wave approximating a one-half sine wave. Time duration is measured between the points where two lines, tangent to the shock wave slopes, intersect the zero g line.

6.3.2 AC Voltage. The expression “AC” wherever used in this specification is defined as 60-cycle, single phase, alternating current.

6.4 Preparation for Delivery Criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

Level A. This level shall be used for those items which are to be shipped to indeterminate designations or stored under indeterminate conditions for redistribution anywhere.

Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

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16.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-T-81290(WP) Test Set, Missile Electrical System Mark 363 Mod 3

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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